

● EASY

● INFORMATION AND IDEAS

Command of Evidence

10 questions · ~15 min

02

Q1

INFORMATION AND IDEAS

Recordings of Female Bottlenose
Dolphins with Their Calves

Dolphin ID	Recording year
FB07	2012
FB25	1989
FB43	1992
FB79	2018

In a study of bottlenose dolphins, biologist Laela S. Sayigh and a team of researchers analyzed recordings of female bottlenose dolphins interacting with their calves.

According to the table, in which year was the dolphin with the ID FB43 recorded with her calf?

- A. 1999
- B. 2012
- C. 2020
- D. 1992

The Minneapolis Institute of Art in Minneapolis, Minnesota, has over 90,000 works of art. Digital images of thousands of those works have been put online through the museum's website and the Google Arts & Culture project. One of the images is of *Mississippi Delta*, a drawing by Siah Armajani. In a paper, a student claims that putting a work from the museum online increases the number of people who experience that work.

Which finding, if true, would most directly support the underlined claim?

- A. The Minneapolis Institute of Art has several works by Siah Armajani.
- B. Many people who have been to Minneapolis say that the Minneapolis Institute of Art is worth visiting.
- C. *Mississippi Delta* has been praised by some art critics.
- D. Each year, more people access the online image of *Mississippi Delta* than visit the Minneapolis Institute of Art in person.

Estimated Impacts of Invasive Predator Species on Threatened Birds, Mammals, and Reptiles

Invasive predator species	Estimated impact on threatened birds	Estimated impact on threatened mammals	Estimated impact on threatened reptiles
Mongoose	0.66	0.75	0.44
Dogs	0.67	0.57	0.44
Cats	0.74	0.61	0.52

How harmful are invasive predators to threatened species? To find out, researchers assigned values from 0.25 to 1.00 to various harms caused by invasive predators. The researchers then used these values to estimate the extent to which several invasive predator species have harmfully impacted threatened birds, mammals, and reptiles. In the table, higher numbers represent greater impacts. The estimated impact of invasive mongooses on threatened mammals is _____blank

Which choice most effectively uses data from the table to complete the example?

- A. 0.35.
- B. 0.61.
- C. 0.75.
- D. 0.52.

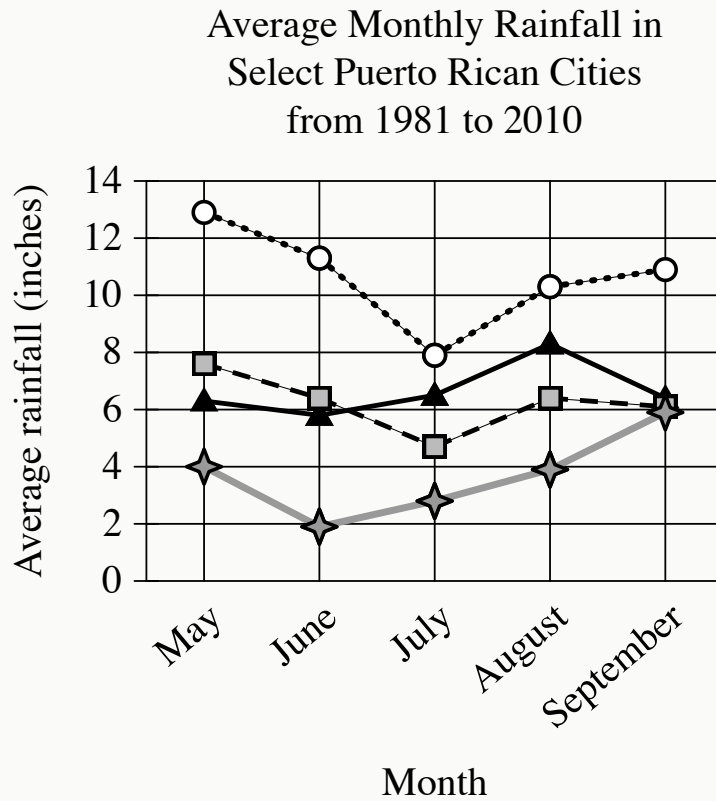
Effect of Paywall Introduction on Newspaper Companies' Revenues

Newspaper	Total revenue change (\$ in thousands)	Percentage change (%)	Newspaper size
<i>Los Angeles Times</i>	93,966	12.5	large
<i>The New York Times</i>	235,788	20	large
<i>The Denver Post</i>	-3,765negative 3,765	-1negative 1	small
<i>Sun Sentinel</i>	-24,899negative 24,899	-11.9negative 11.9	small
<i>Chicago Tribune</i>	94,492	19	large

Digital paywalls restrict access to online content to those with a paid subscription. In an investigation of the effect of paywalls on newspaper company revenues for print and digital subscriptions and advertising, Doug J. Chung and colleagues compared actual outcomes (with a paywall) to control estimates (without a paywall). The researchers concluded that introducing a paywall is generally more beneficial for larger newspapers, which have high circulation and tend to offer a substantial amount of unique online content.

Which choice best describes data from the table that support Chung and colleagues' conclusion?

- A. The *Chicago Tribune* and the *Los Angeles Times* had similar total revenue changes, but the *Los Angeles Times* had a smaller percentage change.
- B. The *Los Angeles Times* had a 12.5% revenue change, while the *Chicago Tribune* had a 19% revenue change.
- C. The *New York Times* had a 20% revenue change, while the *Denver Post* had a –1%negative 1% revenue change.
- D. The *Denver Post* had only a –1%negative 1% revenue change, which was the smallest percentage change of the selected companies.



- The following 4 lines are shown:
 - Rincón
 - Isabela
 - San Sebastián
 - Ponce
- The Rincón line:
 - Begins at May, 6.3 inches
 - Falls gradually to June, 5.8 inches

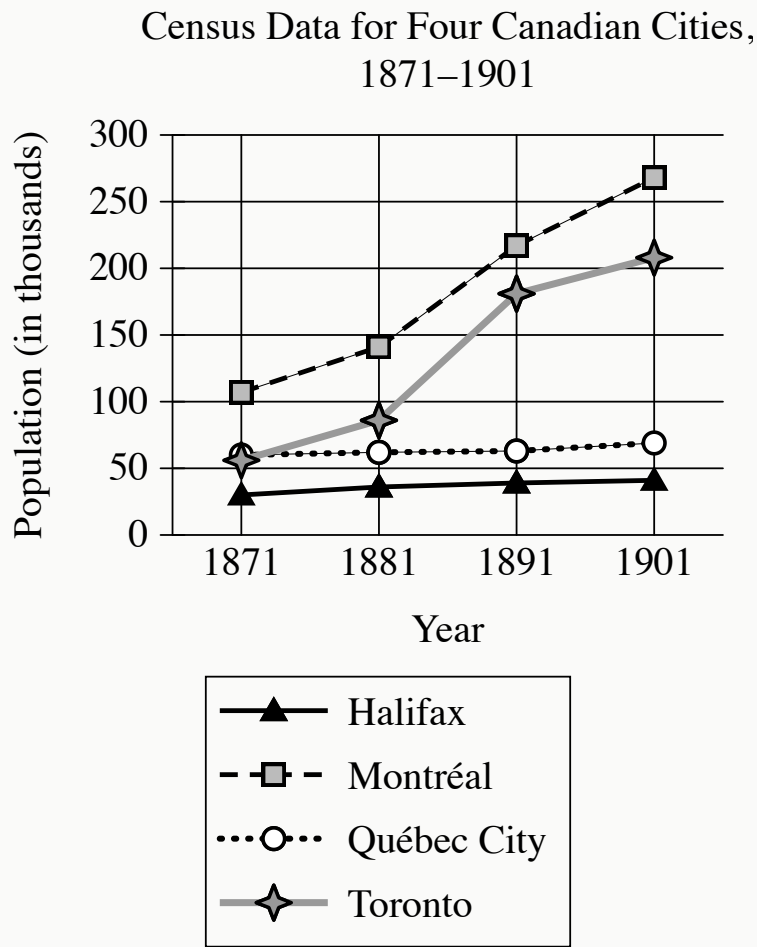
- Rises gradually to July, 6.5 inches
- Rises sharply to August, 8.3 inches
- Falls sharply to September, 6.4 inches
- The Isabela line:
 - Begins at May, 7.6 inches
 - Falls gradually to June, 6.4 inches
 - Falls gradually to July, 4.7 inches
 - Rises gradually to August, 6.4 inches
 - Falls gradually to September, 6.1 inches
- The San Sebastián line:
 - Begins at May, 12.9 inches
 - Falls gradually to June, 11.3 inches
 - Falls sharply to July, 7.9 inches
 - Rises sharply to August, 10.3 inches
 - Rises gradually to September, 10.9 inches
- The Ponce line:
 - Begins at May, 4 inches
 - Falls sharply to June, 1.9 inches
 - Rises gradually to July, 2.8 inches
 - Rises gradually to August, 3.9 inches
 - Rises gradually to September, 5.9 inches

A student is presenting average monthly rainfall totals in various Puerto Rican cities for a science class. During the presentation, the student notes that in September _____ blank

Which choice most effectively uses data from the graph to complete the statement?

- A. Rincón’s average rainfall is greater than that of Isabela, San Sebastián, and Ponce.
- B. Rincón and Ponce have an average rainfall of about 5 inches, and Isabela and San Sebastián have an average rainfall of about 10 inches.
- C. Rincón, Ponce, and Isabela each have an average rainfall below 8 inches, but San Sebastián’s average rainfall that month is greater than 8 inches.

- D. Rincón has a similar average rainfall to Isabela, and Ponce has a similar average rainfall to San Sebastián.



- The following 4 lines are shown:
 - Halifax
 - Montréal
 - Québec City
 - Toronto
- The Halifax line:
 - Begins at 1871, 30 thousand
 - Rises gradually to 1881, 36 thousand
 - Rises gradually to 1891, 39 thousand
 - Rises gradually to 1901, 41 thousand
- The Montréal line:

- Begins at 1871, 107 thousand
- Rises gradually to 1881, 141 thousand
- Rises sharply to 1891, 217 thousand
- Rises sharply to 1901, 268 thousand
- The Québec City line:
 - Begins at 1871, 60 thousand
 - Rises gradually to 1881, 62 thousand
 - Rises gradually to 1891, 63 thousand
 - Rises gradually to 1901, 69 thousand
- The Toronto line:
 - Begins at 1871, 56 thousand
 - Rises gradually to 1881, 86 thousand
 - Rises sharply to 1891, 181 thousand
 - Rises gradually to 1901, 208 thousand

The first national census in Canada was conducted in 1871. The graph shows the population sizes of some of the largest Canadian cities from each of the first four national censuses, which occurred every ten years.

According to the graph, which city had the largest population in 1891?

- A. Toronto
- B. Montréal
- C. Québec City
- D. Halifax

Total Electoral College Votes and Popular Votes in the 15th US
Presidential Election

Candidate	Total electoral college votes	Total popular votes
James K. Polk	170	1,339,494
Henry Clay	105	1,300,004
James Gillespie Birney	0	62,103

A student is writing a paper on the history of the 15th presidential election in the United States, which was held in 1844. The student needs to determine how many electoral college votes were cast for the winning candidate, James K. Polk. According to the table, James K. Polk received _____blank

Which choice most effectively uses data from the table to complete the statement?

- A. 170 electoral college votes.
- B. 103 electoral college votes.
- C. 62 electoral college votes.
- D. 44 electoral college votes.

Effect of Neighboring Species on Pollinator Visits to Target Species

Neighboring species	Target species	Effect value
Virginia spring beauty	star chickweed	0.4853
Himalayan balsam	marsh woundwort	0.7905
common dandelion	cat's ear	-0.6254

Researchers Carolina Laura Morales and Anna Traveset gathered data about flowering plants growing alongside each other in various locations. In each case, the researchers identified one plant as a “target species” and a nearby plant as a “neighboring species.” The researchers then calculated a positive or negative value to show how the neighboring species affected pollinator visits to the target species. One example of a neighboring species with a negative effect value is the _____ blank

Which choice most effectively uses data from the table to complete the example?

- A. Himalayan balsam.
- B. common dandelion.
- C. star chickweed.
- D. Virginia spring beauty.

Hip-hop pedagogy is a form of teaching that's gaining popularity across school subjects. It involves incorporating hip-hop and rap music into lessons as well as using hip-hop elements when teaching other subject matters. For example, Quan Neloms's students look for college-level vocabulary and historical events in rap songs. Researchers claim that in addition to developing students' social justice awareness, hip-hop pedagogy encourages student success by raising students' interest and engagement.

Which finding, if true, would most strongly support the underlined claim?

- A. Students tend to be more enthusiastic about rap music than they are about hip-hop music.
- B. Students who are highly interested in social justice issues typically don't sign up for courses that incorporate hip-hop and rap music.
- C. Educators report that they enjoy teaching courses that involve hip-hop and rap music more than teaching courses that don't.
- D. Courses that incorporate hip-hop and rap music are among the courses with the highest enrollment and attendance rates.

Treasure Island is an 1883 novel by Robert Louis Stevenson. When the narrator was a child, his father ran a hotel. A mysterious sailor came to stay at the hotel. The narrator was frightened of the sailor, as can be seen when the narrator says, _____ blank

Which quotation from *Treasure Island* most effectively illustrates the claim?

- A. “I remember [the sailor] as if it were yesterday, as he came plodding to the inn door, his sea-chest following behind him in a hand-barrow.”
- B. “[The sailor] was a very silent man by custom. All day he hung round the cove or upon the cliffs with a brass telescope.”
- C. “All the time he lived with us [the sailor] made no change whatever in his dress but to buy some stockings from a hawker. One of the [corners] of his hat having fallen down, he let it hang from that day forth, though it was a great annoyance when it blew.”
- D. “How [the sailor] haunted my dreams, I need scarcely tell you. On stormy nights, when the wind shook the four corners of the house and the surf roared along the cove and up the cliffs, I would see him in a thousand forms, and with a thousand diabolical expressions.”